Alignment's Diversity Cost Is Carried by a Small Part of the Shift:
Stage-resolved measurement across four post-training pipelines

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Abstract

Post-training alignment reduces the diversity of language model outputs, and the loss is widely attributed to reinforcement learning on human preferences, with *typicality bias* — a reweighting toward what the base model already found likely — offered as its mechanism. We measure both against four pipelines that publish every intermediate checkpoint: OLMo-2-0425-1B, OLMo-2-1124-7B, OLMo-2-1124-13B and Llama-3.1-Tülu-3-8B. Throughout, *diversity* means next-token entropy at the scored positions.

Composition and consequence come apart. A typicality-aligned component of each stage's log-probability shift accounts for a minority of that shift's variance: a median of at most 2.2% for instruction tuning and 7.1% for the preference stage under an unweighted fit allowing linear and piecewise-linear terms, and at most about 30% under a probability-weighted linear fit. An exact counterfactual Shapley attribution of each stage's entropy change assigns that same component **nearly all (87–139%) of instruction tuning's entropy loss and all (102–291%) of the preference stage's**, under the pre-registered weighted fit. Comparing like with like, a component accounting for about 30% of the shift at most carries nearly all of its entropy cost. The diversity cost is carried by a small part of the shift.

The larger of the two components we measured is a shared direction. Writing ρ_{reg} for the correlation between a stage's per-token shift and the mean shift on *other* prompts (leave-one-prompt-out), and ψ for its correlation with reference log-probability, ρ_{reg} exceeds ψ across the plausible set at every stage of all four models, with intervals excluding zero and null controls at $+0.008$ and $+0.004$. The preference-stage direction promotes dashes, em-dashes, ellipses and brackets and demotes *you* in all four. Neither component is most of the shift: squared, the shared direction accounts for roughly 1–31% of a stage's shift and under 1% for the 1B's instruction tuning, so the majority is explained by neither axis. A second, distinct quantity — how far a stage's entropy falls on aligned-register text, which we call *register-conditional concentration* — transfers across lineage: OLMo-1B's instruction-tuning stage concentrates on text written by OLMo-2-1124-7B-Instruct exactly as much as on its own family's (-0.194 both).

Where you measure changes the answer. Scoring the same checkpoints on the same prompts, Tülu-3-8B's preference stage accounts for 29.2% [22.4, 35.0] of total entropy loss at a chat-template position and 52.8% [50.5, 55.4] over generated continuations. Prompt-final scoring, closest to common practice, gives confidence intervals 4.5–6× wider because it averages one position per prompt rather than ∼190, and on the 1B model it falls 18–28 points below every other convention. We have not found a published stage comparison that reports this choice.

We also report two decoding-time interventions and a correction to our own earlier work. Extrapolating along the base–aligned logit axis recovers diversity at a third of the fluency cost

of temperature at 135M (3.00×, 95% CI [1.58, 5.14]) and does not survive to 1B (1.04×, [0.73, 1.50], $n = 12$, Wilcoxon $p = 0.97$). Reference subtraction, which an earlier draft reported as catastrophic, is not: at matched diversity it sits at 0.98× the temperature frontier, and the earlier figure compared it against $T = 1.0$ instead. Plausible-set ψ never exceeds 0.281 across four models, four stage deltas and three conventions.

1 Introduction

That language models homogenise expression has moved from commentary into measurement. Doshi and Hauser [7] found that writers given LLM story ideas produced work rated more creative individually while the corpus became more similar in aggregate. Sourati, Ziabari and Dehghani [8] synthesised evidence across four disciplines that models reinforce dominant styles.

The technical literature localises the loss to the alignment step. Kirk et al. [4] established a generalisation–diversity trade-off under RLHF, and the phrase "RLHF reduces diversity" is standard. *Typicality bias* [5] supplies a mechanism: human preference data favours the conventional, and KL-regularised optimisation against it amplifies majority-style responses.

Both are usually evaluated by comparing a base model with a released instruct model, collapsing instruction tuning, preference optimisation and any later stages into one delta. Pipelines publishing intermediate checkpoints allow the stages to be separated. We separate them on four, and ask two questions of each stage: what is its shift made of, and which part of it removes entropy. The answers are different components.

1.1 Contributions

1. **Composition is not consequence.** In every model and at both stages, a typicality-aligned component accounting for a minority of the shift's variance carries nearly all of its entropy loss.

2. **A shared-shift geometry, measured and replicated.** $\rho_{\text{reg}} > \psi$ across the plausible set at every stage of all four models, under two continuation sources, with null controls near zero and the token direction stated explicitly. In the probability-weighted region the ordering reverses (Section 4.5).

3. **Cross-lineage transfer of register-conditional concentration**, ruling out checkpoint ancestry though not shared training data.

4. **The measurement convention determines the stage attribution.** Four conventions on four models; the same checkpoints yield preference-stage shares 24 points apart, and the effect is one of statistical precision as well as bias.

5. **Two decoding-time negative results**, one scale-dependent, plus corrections to a matched-diversity error and a withdrawn threshold in our own earlier drafts.

Every claim below is scoped to the models, conventions and regions it was measured on. None is stated as general across post-training pipelines.

2 Measurement

2.1 Models and protocol

pipeline	base	stages	params
OLMo-2-0425	OLMo-2	base → SFT → DPO → Instruct (RLVR)	1B
OLMo-2-1124	OLMo-2	base → SFT → DPO → Instruct (RLVR)	7B
OLMo-2-1124	OLMo-2	base → SFT → DPO → Instruct (RLVR)	13B
Llama-3.1-Tülu-3	Llama 3.1	base → SFT → DPO → Instruct (RLVR)	8B

The 7B and 13B share a release, so comparisons between them vary scale alone. The 1B is a different release (April 2025 vs November 2024), so comparisons involving it hold the recipe near-constant rather than constant.

Diversity, in this paper, is next-token entropy at the scored positions. The generation experiments of Section 6 use mean pairwise Jaccard instead, and say so.

36 prompts across six task families. Each checkpoint is loaded alone, in bf16 with a per-parameter assertion after load, and its next-token log-probability vectors cached before the next is loaded; all delta arithmetic happens afterward on saved arrays. Confidence intervals are bootstrap over prompts, 20,000 resamples.

The stage, register and decomposition analyses ran on Apple MPS; the 1B generation sweep of Section 6.1 and all 135M runs ran on CPU. The Tülu pipeline was reproduced in full on CPU under the prompt-final convention and agrees on every share to within 0.3 percentage points.

Tokeniser identity was verified at the token-ID level for every pipeline, a requirement for logit arithmetic rather than a formality. OLMo needed no restriction. Tülu's SFT, DPO and final heads are 128,264 wide against Llama-3.1-8B's 128,256, so Tülu comparisons use the shared 128,256 IDs, renormalised; excluded mass reaches 7×10^{-6} at prompt-final positions, 1.5×10^{-4} over base-written continuations and 7.8×10^{-4} over final-written ones.

2.2 Four conventions

Entropy and the per-token shift must be evaluated at some set of positions, and the literature does not state which.

- **Raw prompt-final.** One position per prompt: the last token of the raw prompt. Closest to common practice and what our own earlier drafts used.
- **Chat-templated.** The prompt wrapped in the aligned model's chat template, scored at the assistant-start position, with identical token IDs fed to all four stages.
- **Continuation, base-written (primary).** The base model generates 6×32 tokens per prompt; all four stages are teacher-forced through that text so contexts are paired. ∼190 positions per prompt.
- **Continuation, final-model-written.** The same, with the final model as writer.

Base-written continuations were pre-registered as the source for the register test of Section 4 before any of that data existed. They were adopted as primary for the stage shares later, in a dated note appended after the 1B, 7B and 8B continuation shares had been seen but before any 13B data was collected. Both pre-registrations are released with their hashes at `https://github.com/amitd01/axisalignment-paper`, and the appended note is part of the record.

2.3 The convention changes the answer

Preference-stage share of total entropy loss, by convention:

model	prompt-final	chat-template	continuation (base)	continuation (final)
OLMo-2-1B	**−0.6**% [−15.4, 9.1]	25.5% [18.0, 35.2]	27.6% [24.1, 31.2]	17.4% [15.1, 20.1]
OLMo-2-7B	25.7% [19.9, 31.9]	26.1% [21.9, 30.5]	33.3% [30.9, 35.8]	23.9% [22.0, 25.9]
Tülu-3-8B	40.8% [33.0, 50.0]	**29.2%** [22.4, 35.0]	**52.8%** [50.5, 55.4]	44.3% [41.8, 47.0]
OLMo-2-13B	31.4% [22.3, 38.7]	26.8% [22.0, 31.6]	49.0% [46.0, 52.2]	39.7% [37.1, 42.4]

Tülu is the clearest case: 29.2% and 52.8% are the same checkpoints and the same prompts, 24 points apart, and neither position is obviously the wrong one to score.

Prompt-final breaks at small scale. On the 1B model it sits 18–28 points below every other convention, reporting that preference optimisation does nothing measurable — a conclusion no other convention supports. At 7B, 8B and 13B the raw value falls inside or near the range of the others. We observe this at $n = 1$ small model and do not generalise it.

The spread is not monotone in scale: 28.2 points at 1B, 9.4 at 7B, 23.6 at 8B and 22.2 at 13B. The 7B and 13B share a release and differ by more than a factor of two, so sensitivity to position is not a fixed property of a pipeline either.

The effect is about precision as much as bias. Prompt-final averages one position per prompt; continuation scoring averages ∼190. Total-loss confidence intervals are correspondingly 4.5–6× wider by model. Comparing OLMo-7B against Tülu-8B: under prompt-final the totals read 1.123 and 0.963, a paired difference of 0.160 [−0.016, 0.325] — indistinguishable; under continuation scoring they read 0.765 and 0.756, difference 0.009 [−0.038, 0.060] — identical to within ±0.05 nats. The convention moves both the point estimates and how much can be concluded from them.

Whose text matters too, through the denominator. Scoring the final model's own text rather than base-written text lowers the preference share by 8–10 points in every model. This is not the preference stage behaving differently — its absolute drop barely moves (−0.01 to −0.04 nats) — but instruction tuning doing more (+0.07 to +0.19 nats), enlarging the total. The obvious explanation, that the base model is out of distribution on aligned text, is ruled out: base entropy is not higher there but lower at 7B, 8B and 13B (−0.21, −0.27, −0.14) and unchanged at 1B (−0.055 [−0.112, +0.008]).

Reporting standard. State the position convention. Report every stage share beside its absolute drop in nats and the pipeline total, because the three move independently — Section 3 gives a case where a share rises while its absolute cost falls and the total halves.

3 Stage attribution

Under the primary convention:

model	SFT	preference	RLVR	total loss
OLMo-2-1B	**66.1%** [62.2, 69.7]	27.6% [24.1, 31.2]	6.3% [5.0, 7.7]	0.426
OLMo-2-7B	**60.2%** [57.5, 62.8]	33.3% [30.9, 35.8]	6.5% [6.0, 7.0]	0.765
Tülu-3-8B	40.7% [37.9, 43.2]	**52.8%** [50.5, 55.4]	6.5% [5.8, 7.3]	0.756
OLMo-2-13B	40.5% [36.7, 44.1]	**49.0%** [46.0, 52.2]	10.5% [9.8, 11.4]	0.413

Which stage is larger varies by model, and within the OLMo-1124 release it flips between the 7B and the 13B. The universal form of "RLHF reduces diversity" is not supported, and neither is its negation.

A share is not a cost. Between the 7B and 13B — same release, scale the only variable — the preference share rises from 33.3% to 49.0%, a paired difference of +15.6 points [+12.7, +18.6]. Over the same step its *absolute* entropy cost falls, 0.255 → 0.202 nats (−0.053 [−0.078, −0.028]), and the pipeline total nearly halves, 0.765 → 0.413, because instruction tuning's cost collapses from 0.460 to 0.167. Under the chat-template convention nothing changes at all (+0.7 [−3.6, +4.7]). "The preference share rises with scale" is true here and means close to the opposite of what it sounds like.

Total collapse is non-monotone within OLMo (1B < 7B > 13B under both prompt-final and continuation scoring). The 13B loses the least of any model under prompt-final (0.469 against 0.602 for the next lowest), but under continuation scoring it is indistinguishable from the 1B (−0.013 [−0.053, +0.027]), and under the chat template it loses more. There is no scale trend in total collapse across these four.

RLVR is small everywhere: 3.6–10.5% across all models and conventions, though the 7B and 13B differ within the primary convention (6.5% [6.0, 7.0] against 10.5% [9.8, 11.4]).

4 What the shift is made of

4.1 Two geometries

For each stage delta we compute the per-token shift $\delta(v) = \log p_{\text{target}}(v \mid c) - \log p_{\text{ref}}(v \mid c)$ and correlate it with two things:

- ψ — the reference model's own log-probability, $\log p_{\text{ref}}(v \mid c)$. This is the quantity the typicality account makes a claim about.
- ρ_{reg} — the mean shift at all positions of the *other* 35 prompts (leave-one-prompt-out, so no in-sample leakage). High ρ_{reg} means the stage applies the same shift regardless of context.

We also report the partial correlation of δ with the shared direction controlling for $\log p_{\text{ref}}$, which asks whether the shared direction is typicality under another name.

4.2 The shared shift exceeds typicality across the plausible set

Pre-registered test, base → SFT, plausible region ($p_{\text{ref}} > 10^{-6}$), base-written continuations:

model	ψ	ρ_{reg}	partial	$\rho_{\text{reg}} - \psi$
OLMo-2-1B	-0.017	$+\mathbf{0.092}$ [+0.083, +0.101]	+0.089	+0.109 [+0.098, +0.120]
OLMo-2-7B	+0.082	$+\mathbf{0.320}$ [+0.310, +0.331]	+0.324	+0.238 [+0.222, +0.253]
Tülu-3-8B	+0.069	$+\mathbf{0.263}$ [+0.248, +0.276]	+0.267	+0.194 [+0.178, +0.208]
OLMo-2-13B	+0.118	$+\mathbf{0.445}$ [+0.431, +0.457]	+0.447	+0.326 [+0.307, +0.345]

All four pass the pre-registered criterion: $\rho_{\text{reg}} > 0$, $\rho_{\text{reg}} > \psi$, partial > 0, all intervals excluding zero, under both continuation sources (the smallest margin under final-written text is 0.097 [0.085, 0.111], at the 13B preference delta). **Null controls** — identical weights evaluated on two numerical backends, run on OLMo-1B and Tülu-8B — give $\rho_{\text{reg}} = +0.008$ [+0.003, +0.014] and +0.004 [−0.000, +0.009]. The partial correlation is within 0.003 of, or larger than, the raw one in every case, so the shared direction is not typicality in disguise.

The preference delta shows the same pattern more strongly at the 1B, 7B and Tülu ($\rho_{\text{reg}} =$ +0.558, +0.489, +0.486) and less so at the 13B, where it is +0.386 — smaller than that model's base → SFT value, with $\rho_{\text{reg}} - \psi$ shrinking from 0.326 to 0.130.

4.3 The direction, in tokens

The preference-stage shared shift promotes dashes, em-dashes, ellipses and brackets, and demotes *you*, in all four models and under both continuation sources. *I* is among the fifteen most-demoted tokens in the 1B, 7B and Tülu under base-written text, but not in the 13B under either source, nor in Tülu under final-written text. In the 1B and 7B the shift also demotes newlines and end-of-sequence — which instruction tuning had *promoted* in those same models — while Tülu and the 13B promote paragraph-break tokens instead. Hedges (*effectively*, *essentially*, *clearly*) are top-ranked only in the 1B.

Token lists are per delta and per model; the stable core across all four is punctuation up, second person down.

4.4 A related quantity transfers across lineage

This subsection concerns a different quantity from the preceding one. **Register-conditional concentration** is how much further a stage's entropy falls when scoring aligned-register text rather than base-written text; **shared shift** (ρ_{reg}) is the context-invariance of the log-probability shift. The lineage test below measures the former, not the latter.

Register-conditional concentration could be nothing more than a model being sharper on text written by its own descendants. OLMo-1B and OLMo-2-1124-7B share a tokeniser, which permits a direct test: score 7B-Instruct-written text through the 1B stages, where no 1B checkpoint is an ancestor of the writer.

The concentration is identical: $\mathbf{X}[\text{SFT}] = \mathbf{-0.194}$ [−0.224, −0.163] against same-lineage −0.194 [−0.230, −0.158]. Later stages add nothing beyond SFT.

This rules out checkpoint ancestry as the explanation for register-conditional concentration. It says nothing directly about whether the shared direction transfers; we did not test that. It does not rule out shared training data — both pipelines use Tülu-3-style instruction mixtures — and that remains the live alternative explanation.

4.5 At the head of the distribution it looks different

The picture above is of the broad plausible set, with a median of roughly 900–3,800 tokens per position. In the probability-weighted region the ordering reverses: ψ_w exceeds ρ_w for base → SFT

in all four models, and for the preference delta at 7B, 8B and 13B (the 13B reads ψ_w 0.433 against ρ_w 0.096; OLMo-7B's base → SFT gives $\rho_w - \psi_w = -0.203$). OLMo-1B's preference delta is the exception, with ρ_w 0.435 > ψ_w 0.163.

The probability-weighted statistic is heavily concentrated: the tokens covering 50% of base mass number a *median of two*. So "the head looks like reweighting" is a statement about the top one or two candidates per position, not about half the distribution. Both readings hold at their own scope.

ψ in absolute terms. Across four models and four stage deltas, plausible-set ψ never exceeds **0.281** under prompt-final and the two continuation conventions, and never exceeds 0.256 under the primary convention. Probability-weighted, it never exceeds 0.566 under those three; including chat-template positions it reaches 0.663 (13B, DPO → final), whose plausible-set maximum is 0.237.

5 What costs the diversity

Composition and consequence are different questions, and this section separates them. We attribute each stage's entropy change causally using counterfactual distributions — apply only part of the stage's log-probability shift, renormalise, recompute entropy — with Shapley values over the parts, which sum exactly to the measured ΔH with no residual interaction term.

5.1 A component explaining a minority of the shift carries nearly all of the loss

The pre-registered result. Under the weighted linear fit specified in advance, the typicality-aligned component carries **nearly all of instruction tuning's entropy loss** (87.0% [84.9, 88.9] at the 13B, 94.5% [92.7, 96.4] at the 7B, 96.6% [94.5, 98.8] at Tülu, 139.0% [131.7, 147.5] at the 1B) **and all of the preference stage's** (102.2% to 290.5%). Only the preference stage and the 1B's instruction tuning reach 100%; the other three instruction-tuning intervals sit below it. Shares above 100% with opposite-signed partners mean the components interact; magnitudes are not separable there, but the sign and the ordering are.

How much of the shift that component is — exploratory. The variance analysis below was run after the other results, to ask how large the costly component is. It was not pre-registered and is reported as exploratory. Median R^2 of each stage's shift over the plausible set explained by typicality (linear and piecewise-linear in reference log-probability), base-written continuations, 400 sampled positions per model:

model	base → SFT	SFT → DPO
OLMo-2-1B	0.7%	1.0%
OLMo-2-7B	1.5%	3.6%
Tülu-3-8B	1.4%	6.2%
OLMo-2-13B	2.2%	7.1%

Recomputed over all ∼6,800 positions per model with bootstrap intervals, these hold: 0.73% [0.68, 0.76], 1.57% [1.37, 1.78], 1.45% [1.29, 1.64] and 2.20% [1.80, 2.58] for instruction tuning across the 1B, 7B, Tülu and 13B, and 1.1–7.1% for the preference stage with an upper bound of 7.9%. Under a probability-weighted *linear* fit — the pre-registered one, with no piecewise-linear term — the same quantity is larger: up to 19.2% [16.9, 21.4] for instruction tuning and 30.4% [27.96, 33.42]

for the preference stage. An exploratory variant merging linear and non-linear typicality assigns the component 53–122% and 95–204% of the two stages' losses.

Comparing like with like — the weighted linear fit on both sides — a component accounting for about 30% of a stage's shift at most carries nearly all of its entropy loss, in every model and at both stages. Under an unweighted fit admitting a piecewise-linear term, the same component accounts for at most 7.1%. That is the paper's central measurement, and the weighted pairing is the conservative one.

The remainder is not inert. For the preference stage, the component orthogonal to both axes falls with scale within the OLMo-1124 release: 25.7% [22.1, 29.3] of the entropy loss at the 7B against 7.4% [5.2, 9.7] at the 13B, with Tülu-8B — a different family — at 14.7% [12.2, 17.5]. The 1B's 45.8% [37.0, 57.0] is not comparable to those: its decomposition nearly cancels there (typicality 290.5% against a shared share of −236.2%), so only the positive sign is interpretable.

At instruction tuning the 1B is the interesting case, and it runs the other way: the remainder *raises* entropy, −30.6% [−37.7, −24.1], partly offsetting a typicality component of 139.0%. That decomposition does not near-cancel — the shared share is 2.1% — so unlike the 1B preference stage its magnitude can be read. At the 7B, Tülu and 13B the remainder carries 1–6%.

Attributing nearly all of the loss to the typicality-aligned component is therefore a statement about that component's share, not a demonstration that nothing else moves entropy. We do not characterise what the remainder is.

5.2 Whether the two stages differ depends on the fit

The preference stage. Its entropy loss comes mostly from the body of the distribution: the head (reference-top tokens covering 50% of mass, median 2 tokens) carries 15–23%, the body 74–86%. Decomposing by component, the loss is carried by the typicality-aligned part while the shared-shift part pushes entropy back *up* — by 16%, 28% and 55% at the 13B, Tülu-8B and OLMo-7B on the weighted fit (18%, 28%, 47% unweighted). The weighted typicality-minus-shared contrast runs +118 to +527 points with intervals clear of zero in all four models.

Sampling 400 positions each **at the 1B, 7B and Tülu-8B** (not the 13B), the typicality-aligned component is a positive sharpening at 85–97% of positions, with weighted median $\beta \approx +0.09$ to $+0.12$ — a local temperature of about 0.9. The 1B is the exception **for the preference stage**: $\beta \approx 0$ there, with components nearly cancelling, so only their signs are interpretable. Its instruction-tuning stage behaves like the others, $\beta \approx +0.10$. β was not measured at the 13B, where only the Shapley shares are available.

Instruction tuning, and a fit-dependence we initially mistook for a mechanism. An earlier draft reported that instruction tuning's costly component was not a temperature while the preference stage's was. That contrast is an artifact of comparing the two stages under different fits.

Under the **weighted** fit — the pre-registered one — the two stages look alike. Instruction tuning's sharpening coefficient is $\beta \approx +0.08$ to $+0.115$, against the preference stage's $+0.09$ to $+0.12$, and linear typicality alone carries 87–139% of instruction tuning's loss.

Under an **unweighted** fit they diverge. A single linear coefficient then captures between −29% and 51% of instruction tuning's loss, and a piecewise-linear function of reference log-probability is needed: it captures 200%, 54%, 76% and 23% across the four models on the pre-registered statistic, or 122% / 56% / 67% / 53% merging linear and non-linear typicality (they are collinear, so the split between them is arbitrary). The 7B and 13B interval lower bounds are 49% and 41%.

So: **under an unweighted fit instruction tuning's loss needs a non-linear term and the preference stage's does not; under a weighted fit both look like a local sharpening of similar size.** We report both and claim no mechanism difference between the stages.

A further earlier claim — that no decoding-time operator built on the reference axis could reach instruction tuning's loss — is **not supported**. On the pre-registered non-linear statistic it is too strong at the 1B and at Tülu, unresolved at the 7B, and holds only at the 13B. A non-linear, truncation-like reference operator is not ruled out.

6 Two decoding-time interventions

6.1 Extrapolating along the base–aligned axis

DeRa [3] proves that aligned models at varying KL strength are geometric mixtures differing only in mixing weight:

$$z = z_b + \lambda(z_a - z_b)$$

$\lambda = 1$ recovers the aligned model, $\lambda = 0$ the base model, $\lambda < 0$ extrapolates past the base model away from alignment. Published evaluations cover $\lambda \in [0, 2]$.

Diversity in this section is mean pairwise Jaccard, and fluency is median GPT-2 perplexity — an independent judge sharing neither family nor tokeniser with either model in the mixture.

At **135M** (SmolLM2 base/instruct), the per-prompt paired comparison at matched diversity gives $\lambda = -1.0$ beating temperature by **3.00×, 95% CI [1.58, 5.14]**, winning on 4/5 prompts. Re-reading the published scored numbers at matched diversity gives $\lambda < 0$ ratios of 0.17–0.37 against the temperature frontier.

At **1B** it does not survive. With the full design of 12 prompts, the paired comparison gives **1.04×, 95% CI [0.73, 1.50]**, λ winning on 5/12, Wilcoxon $p = 0.970$ against an attainable floor of 0.0005. The estimate declines monotonically from $n = 7$ onward: 1.16×, 1.08×, 1.04×. The interval excludes the 135M effect size, whose lower bound is 1.58, but does not exclude a modest effect up to 1.5×.

A direction control — the delta shuffled across the vocabulary at each step, preserving its norm and the full marginal distribution of its values — is decisive at 135M (401.3 against temperature's 403.1 at matched diversity, while the true delta gives 165.4) and **uninformative at 1B**, where both arms have degenerated into the thousands by the time their diversity matches.

6.2 Reference subtraction, and a correction

Writing $p_\gamma \propto p_{\text{align}} \cdot p_{\text{ref}}^{-\gamma}$ attenuates the reference's contribution — the repair the typicality account recommends, and contrastive decoding [1] with the base model as amateur.

An earlier draft of this work reported that $\gamma = 0.5$ raised judge perplexity to 19,956 and treated this as catastrophic failure. **That comparison was against $T = 1.0$, not at matched diversity.** Re-read at matched diversity, $\gamma = 0.5$ sits at **0.98× the temperature frontier** at its own Jaccard. It overshot along the frontier; it did no off-frontier harm. The original matched-diversity table stopped short of that diversity level, so the condition was never properly compared.

A pre-registered sweep $\gamma \in \{0.05, 0.1, 0.2, 0.3, 0.5\}$ on the 135M pair finds nothing degenerate at small γ. $\gamma = 0.05$ is classified temperature-like (1.21×, CI [0.79, 1.33]) but fragile — only 52% of bootstrap draws fall inside the temperature curve's interpolation range; $\gamma = 0.1$ misses the band by 0.03. With five prompts the sweep neither falsifies nor establishes the hypothesis.

The measured β on that pair is $+0.081$ at continuation positions. That is the *whole* base $\rightarrow$ Instruct pipeline, since SmolLM2 publishes no intermediate checkpoints, so it is comparable in size to the preference-stage β above but is not the same quantity.

6.3 A partial explanation

For the preference stage at 7–8B, the decomposition suggests why an operator on the reference axis would track the temperature frontier rather than beat it: the entropy-relevant part of that stage's shift is itself close to a temperature.

This cannot be the explanation for the 1B null. There the preference stage's $\beta \approx 0$, and λ acts on the whole base $\rightarrow$ Instruct delta, which at the 1B is 66% instruction tuning — whose costly component looks non-linear under an unweighted fit and temperature-like under a weighted one. We offer the account as a hypothesis for 7–13B and note that the 1B result is unexplained.

7 What we got wrong

Four earlier conclusions in this work were overturned, by four different causes. We list them with their causes because a vague account of one's own errors is worth less than a precise one.

earlier claim	what overturned it
"Instruction tuning carries ∼99% of the entropy loss"	position convention
A 1B $\rightarrow$ 7B jump in the preference share	position convention
"Reference subtraction fails catastrophically"	comparison baseline ($T = 1.0$ instead of matched diversity)
"No reference-axis operator can reach SFT's loss"	operationalisation (linear-only fit)

Two further claims fell to ordinary causes: "instruction tuning does all of it" as a *general* claim was refuted by measuring a second pipeline, and a reported negative sign on the 1B preference-delta ψ did not survive a change of numerical backend.

Two analytical predictions also failed, in opposite directions.

$\lambda < 0$ was predicted catastrophic. An analytical model predicted $\lambda < 0$ would be harmful at any scale, because λ rescales the whole reward and reversing it reverses the quality term with the typicality term. At 135M, generation showed the opposite. The model treated the reference as an abstract distribution over items with independently drawn quality; real base models encode grammaticality, so both endpoints of the λ axis are fluent language models and the segment between them lies inside the manifold of well-formed text.

A ψ threshold was derived and does not hold. The same model was used to derive a crossover $\psi^* \approx 0.71$–0.83 below which reference subtraction should lose to temperature, and three earlier drafts compared measured ψ against it. ψ^* is not a constant: holding the model's other parameters fixed and varying only the matched-diversity target moves it from 0.000 to 1.000, and the published value is the most demanding point on that curve. The model is also off-regime — its aligned distribution collapses 47-fold against its reference, where the measured collapse is 1.5–2.2$\times$ under continuation scoring (1.51–3.22$\times$ under prompt-final) — and rebuilt at measured peakedness and collapse it yields ψ^* between 0.000 and 0.424, below several measured values. **We withdraw**

the threshold and make no decision-theoretic claim. ψ is reported as an absolute magnitude throughout.

The assumption behind both failures — quality independent of typicality — is common in this literature and reverses conclusions.

8 Related work

Decoding-time model composition is mature and has been used almost exclusively to add capability. Contrastive decoding [1] maximises expert-minus-amateur log-likelihood to suppress generic output. Proxy-tuning [2] adds a small-model tuning delta to a large base model. DeRa [3] proves the geometric-mixture structure and sweeps $\lambda \in [0, 2]$. None evaluates against diversity metrics and none explores $\lambda < 0$.

The typicality-bias literature proposes prompt-level interventions: verbalized sampling, diverse personas, chain-of-thought differentiation [5]. These require the user to want diversity, which matters given that the collective-diversity loss Doshi and Hauser [7] identify is a social dilemma no individual has an incentive to solve. Related work on preference collapse [6] shows minority preferences being virtually disregarded under standard alignment objectives.

Stage-resolved analysis. Kirk et al. [4] compare SFT against RLHF on generalisation and diversity but do not decompose a single pipeline's entropy budget across its stages. We have not found a cross-pipeline comparison of such a decomposition, nor a stage comparison that reports the position convention of Section 2.3.

9 Limitations

1. **Four pipelines, two base families.** The shared-shift result replicates within two lineages that use related instruction mixtures.
2. **The lineage test rules out ancestry, not data.** Both families train on Tülu-3-style mixtures, so shared training data remains a live explanation.
3. β **was measured at the 1B, 7B and Tülu-8B**, on 400 sampled positions, but not at the 13B. The 1B preference stage is the one place it is ≈ 0. The R^2 and Shapley results in Section 5.1 cover all four models.
4. **"Typicality" is operationalised** as a linear and piecewise-linear function of reference log-probability. It is one operationalisation of a claim the source literature does not state formally. Section 5.2 shows that two analyst choices — linear versus non-linear, and weighted versus unweighted — each reverse a conclusion.
5. **One judge.** GPT-2 perplexity measures fluency, not whether an answer is good, and it rewards predictable text, biasing against diverse conditions. Human evaluation is absent and is the limitation we would most want addressed.
6. **The γ sweep is underpowered** at five prompts and resolves nothing.
7. **The 1B λ null is unexplained.**
8. **Scale.** 135M, 1B, 7B, 8B, 13B. Nothing here speaks to frontier scale.

10 Conclusion

Across four pipelines, alignment's diversity cost is carried by a small part of the alignment shift. A typicality-aligned component accounting for about 30% of a stage's shift at most — and under an unweighted fit, at most 7% — carries nearly all of its entropy loss, at both stages, in every model measured. Larger than that component, though itself a minority of the shift, is a shared direction: measurable at every stage of every model, surviving a partial correlation against reference typicality, with null controls at 0.008 and 0.004. Its context-invariance depends on stage and scale together, and in opposite directions: within OLMo, instruction tuning rises with scale (0.092, 0.320, 0.445 at 1B, 7B and 13B) while the preference stage falls (0.558, 0.489, 0.386). It is moderately context-invariant everywhere — ρ_{reg} between 0.26 and 0.56 — except for instruction tuning at the 1B. Squared, that is at most roughly 0.31 of the shift — the square of a mean correlation, so an approximation rather than a bound — and it is not a context-independent direction so much as a partially shared one. Most of each stage's shift is accounted for by neither axis, we do not characterise what the remainder is, and it is not inert — at instruction tuning in the 1B it raises entropy by enough to offset a third of the loss (Section 5.1).

Those findings suggest a hypothesis about larger scales, which this work does not test: an operator on the reference axis acts on the axis that carries the entropy, and at 7–8B the entropy-relevant part of the preference shift is already close to a temperature, so there would be little for such an operator to recover that temperature does not. It is a hypothesis and not an account of what we measured. No decoding-time intervention was run above 1B. It cannot be the explanation of the 1B null, where the preference stage's $\beta \approx 0$ and λ acts mostly on instruction tuning (Section 6.3); that null stays unexplained, and the same operator beat temperature threefold at 135M. The results also raise a question this work does not answer. The field measures alignment's effect through diversity metrics, and most of what alignment does to the next-token distribution does not appear in them.

Underneath sits a methodological result we did not expect to be consequential. The same Tülu-3-8B checkpoints yield preference-stage shares of 29% and 53% depending on where the distribution is scored, and the 1B model's conventional measurement sits 18–28 points from every alternative. Two of this work's own earlier conclusions were reversed by that choice alone. Stage attribution is not reproducible without stating it.

A Reproduction

file	contents
`dump_logprobs_mps.py`	per-checkpoint log-prob dumps
`analyse_psi_family.py`	stage-resolved ψ and entropy, bootstrap over prompts
`register_v2.py`	shared-shift geometry, both arms, null controls
`decomp.py`	counterfactual Shapley entropy decomposition
`decomp_resid.py`	non-linear typicality, cross-fitted shared directions
`check_tokenizers.py`	token-ID-level tokeniser identity verification
`grind.py`, `matched6.py`	generation sweep and matched-diversity scoring
`gen_scale.py` (mode `ctrl`)	shuffled-delta direction control
`exp1e_peakedness.py`	the probe that withdrew the ψ^* threshold (Sec. 7)
`pilot135m_port/`	135M γ sweep and β measurement, ported
`pilot-135m/`	the original 135M pilot, including the withdrawn threshold and the source of the 3.00× figure
`logs/`	run scripts and evaluation outputs behind the reported numbers
`logs/inline_checks.py`	reproduces the figures originally computed inline rather than by a script

Six pre-registrations, each hashed before its data existed, are released with the code at `https://github.com/amitd01/axisalignment-paper`: the 13B scale test, the register test, the lineage test, the entropy decomposition, the residual test and the 135M γ sweep. Two carry dated notes appended after the fact; the hashes cover the original text and the notes state what changed and when.

Seeds: 20261012 (1B), 20260803 (135M). $N = 6$ samples, 32 new tokens, top-p 0.95 held constant. Fixed seeds do not guarantee bit-identical output across hardware [11]; batch-invariant kernels were not attempted.

Consistency check. $T = 1.0$, $\lambda = +1.0$ and $\gamma = 0$ are the same distribution by construction and returned identical metrics on the original cloud run (1B: Jaccard 0.0746, ppl 58.8). It was not re-run on the hardware used for Sections 3–5.

Metrics. Entropy is the diversity measure throughout Sections 2–5. Section 6 uses mean pairwise Jaccard; distinct-n saturates at 1.000 by $T = 1.6$ at these scales and is reported only for comparability with prior work. Perplexity is a median across samples.

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